Assign 4

과제 풀이 정리 · Notion 아카이브

Problem 3 : *ModelBasedMonteCarlo*

3a : Value Iteration

```
def valueIteration(
    succAndRewardProb: Dict[
        Tuple[StateT, ActionT],
        List[Tuple[StateT, float, float]]
    ],
    discount: float,
    epsilon: float = 0.001
) -> Dict[StateT, ActionT]:
    stateActions = defaultdict(set)
    for state, action in succAndRewardProb.keys():
        stateActions[state].add(action)
    if not stateActions:
        return {}
    def computeQ(
        V: Dict[StateT, float],
        state: StateT,
        action: ActionT
    ) -> float:
        ret = 0
        for next_state, prob, reward in succAndRewardProb[(state, action)]:
            ret += prob * (reward + discount * V[next_state])
        return ret
    def computePolicy(
        V: Dict[StateT, float]
    ) -> Dict[StateT, ActionT]:
        policy = {}
        for state in stateActions:
            policy[state] = max(
                stateActions[state],
                key=lambda a: (computeQ(V, state, a), a)
            )
```

```
    return policy
print('Running valueIteration...')
V = defaultdict(float)
numIters = 0
while True:
    newV = defaultdict(float)
    for state in stateActions:
        newV[state] = max(
            computeQ(V, state, action)
            for action in stateActions[state]
        )
    delta = max(
        abs(V[state] - newV[state])
        for state in stateActions
    )
    V = newV
    numIters += 1
    if delta <= epsilon:
        break
V_opt = V
print(("valueIteration: %d iterations" % numIters))
return computePolicy(V_opt)
```

3b÷ Model-Based Monte Carlo

Model-Based Monte Carlo의 핵심은 환경의 전이 확률과 보상을 직접 알지 못해도, 경험으로 추정한 모델을 이용해 최적 정책을 구할 수 있다는 점이다. 먼저 (state, action) 쌍마다 다음 상태로의 전이 횟수와 보상의 합을 기록한다. 전이 확률은 해당 (state, action)에서 관측된 전이 횟수를 전체 전이 횟수로 나눈 값이고, 평균 보상은 해당 전이에서 관측된 보상의 합을 전이 횟수로 나눈 값이다. 이렇게 추정한 전이 확률과 평균 보상을 이용해 succAndRewardProb를 구성한 뒤, 일정 간격마다 valueIteration을 호출해 최적 정책을 다시 계산한다.

```
class ModelBasedMonteCarlo(util.RLAlgorithm):
    def __init__(
        self,
        actions: List[ActionT],
        discount: float,
        calcValIterEvery: int = 10000,
        explorationProb: float = 0.2,
    ) -> None:
```

```
    self.actions = actions
    self.discount = discount
    self.calcValIterEvery = calcValIterEvery
    self.explorationProb = explorationProb
    self.numIters = 0
    # (state, action) -> {nextState -> ct} for all nextState
    self.tCounts = defaultdict(
        lambda: defaultdict(int)
    )
    # (state, action) -> {nextState -> totalReward} for all nextState
    self.rTotal = defaultdict(
        lambda: defaultdict(float)
    )
    self.pi = {}  # Optimal policy for each state. state -> action

def getAction(
    self,
    state: StateT,
    explore: bool = True
) -> ActionT:
    if explore:
        self.numIters += 1
        explorationProb = self.explorationProb
        if self.numIters < 2e4:
            explorationProb = 1.0
        elif self.numIters > 1e6:
            explorationProb = (
                explorationProb
                / math.log(self.numIters - 100000 + 1)
            )
    if (
        (explore and random.random() < explorationProb)
        or (state not in self.pi)
    ):
        return random.choice(self.actions)
    else:
        return self.pi[state]
```

```
  def incorporateFeedback(
    self,
    state: StateT,
    action: ActionT,
    reward: int,
    nextState: StateT,
    terminal: bool
  ):
    self.tCounts[(state, action)][nextState] += 1
    self.rTotal[(state, action)][nextState] += reward
    # state, action 쌍의 전이 횟수와 보상 합계를 기록한다.
    if self.numIters % self.calcValIterEvery == 0:
      succAndRewardProb = defaultdict(list)
      for (s, a), next_states in self.tCounts.items():
        total_trans = sum(next_states.values())
        for ns, count in next_states.items():
          prob = count / total_trans
          reward = self.rTotal[(s, a)][ns] / count
          succAndRewardProb[(s, a)].append(
            (ns, prob, reward)
          )
      self.pi = valueIteration(
        succAndRewardProb,
        self.discount
      )
```

Problem 4 : *Model-free RL*

4a : TabularQ-Learning(큐러닝)

```
class TabularQLearning(util.RLAlgorithm):
  def __init__(
    self,
    actions: List[ActionT],
    discount: float,
    explorationProb: float = 0.2,
    initialQ: float = 0
  ):
    self.actions = actions
    self.discount = discount
```

```
    self.explorationProb = explorationProb
    self.Q = defaultdict(lambda: initialQ)
    self.numIters = 0

def getAction(
    self,
    state: StateT,
    explore: bool = True
) -> ActionT:
    if explore:
        self.numIters += 1
        explorationProb = self.explorationProb
        if self.numIters < 2e4:
            explorationProb = 1.0
        elif self.numIters > 1e5:
            explorationProb = (
                explorationProb
                / math.log(self.numIters - 100000 + 1)
            )
    if (
        explore and random.random() < explorationProb
    ):
        return random.choice(self.actions)
    else:
        max_q = max(
            self.Q[state, a]
            for a in self.actions
        )
        best_actions = [
            a
            for a in self.actions
            if self.Q[state, a] == max_q
        ]
        return random.choice(best_actions)

def getStepSize(self) -> float:
    return 0.1
```

```
    def incorporateFeedback(
        self,
        state: StateT,
        action: ActionT,
        reward: float,
        nextState: StateT,
        terminal: bool
    ) -> None:
        a = self.getStepSize()
        if terminal:
            target_V = reward
        else:
            target_V = (
                reward
                + self.discount * max(
                    [
                        self.Q[nextState, act]
                        for act in self.actions
                    ]
                )
            )
        self.Q[state, action] = (
            (1 - a) * self.Q[state, action]
            + a * target_V
        )
```

4b÷ FourierFeature(특징)extractor

```
def fourierFeatureExtractor(
    state: StateT,
    maxCoeff: int = 5,  # maxCoeff는 푸리에 계수 k의 최댓값
    scale: Optional[Iterable] = None
) -> np.ndarray:
    if scale is None:
        scale = np.ones_like(state)
    features = None
    scaled_state = np.array(state) * np.array(scale)
    coeff_list = np.arange(maxCoeff + 1)  # [0, 1, 2, ..., maxCoeff]
    for s_s in scaled_state:
```

```
        multiples = s_s * coeff_list
        if features is None:
            features = multiples
        else:
            features = np.add.outer(features, multiples).flatten()
    features = np.cos(np.pi * features)
    return features
```

4c÷Q-learning with Function Approximation

```
class FunctionApproxQLearning(util.RLAlgorithm):
    def __init__(
        self,
        featureDim: int,
        featureExtractor: Callable,
        actions: List[int],
        discount: float,
        explorationProb=0.2
    ):
        self.featureDim = featureDim
        self.featureExtractor = featureExtractor
        self.actions = actions
        self.discount = discount
        self.explorationProb = explorationProb
        self.W = np.random.standard_normal(
            size=(featureDim, len(actions))
        )
        self.numIters = 0

    def getQ(
        self,
        state: np.ndarray,
        action: int
    ) -> float:
        features = self.featureExtractor(state)
        weight = self.W[:, action]
        return np.dot(weight, features)

    def getAction(
```

```
    self,
    state: np.ndarray,
    explore: bool = True
) -> int:
    if explore:
        self.numIters += 1
        explorationProb = self.explorationProb
        if self.numIters < 2e4:
            explorationProb = 1.0
        elif self.numIters > 1e5:
            explorationProb = (
                explorationProb
                / math.log(self.numIters - 100000 + 1)
            )
    if (
        explore and random.random() < explorationProb
    ):
        return random.choice(self.actions)
    else:
        max_q = max(
            self.getQ(state, a)
            for a in self.actions
        )
        best_actions = [
            a
            for a in self.actions
            if self.getQ(state, a) == max_q
        ]
        return random.choice(best_actions)

def getStepSize(self) -> float:
    return 0.005 * (0.99) ** (
        self.numIters / 500
    )

def incorporateFeedback(
    self,
    state: np.ndarray,
```

```
    action: int,
    reward: float,
    nextState: np.ndarray,
    terminal: bool
) -> None:
    a = self.getStepSize()
    if terminal:
        target = reward
    else:
        V_opt = max(
            [
                self.getQ(nextState, act)
                for act in self.actions
            ]
        )
        target = reward + self.discount * V_opt
    features = self.featureExtractor(state)
    expect = self.getQ(state, action)
    self.W[:, action] = (
        self.W[:, action]
        + a * (target - expect) * features
    )
```

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